1. PROTECTED RAW INPUT AND NO-BLINK HOLD-UP
RAW_POWER_FROM_PWR_SELECT
Mates with PWR-SELECT J301; 15 A path; do not mate live.
4 x 5600 uF installed = 22.4 mF. PWR-SELECT local 660 uF remains additive.
Stuff C1104-C1108 only after source MOSFET SOA and inrush tests; fully stuffed total is about 47.7 mF.
2. SYS_5V15 / 12 A SYNCHRONOUS BUCK
TI 5 V / 12 A reference adapted to 5.144 V. 300 kHz; target loop crossover about 20 kHz.
3. AUX_12V / 8 A FOUR-SWITCH BUCK-BOOST
Revised 12 V / 8 A starting design. Recalculate losses and compensation, then verify by Bode plot and full thermal load before routing.
4. DEDICATED RADIO, NETWORK, AND LOGIC BUCKS
TPS259827L active current limiter; 255 ohm ILIM; 1320 uF local modem bulk; validate weak-signal TX bursts.
TPS22990 provides controlled Wi-Fi startup and PG; it is not a current limiter.
5. LOW-NOISE POINT-OF-LOAD RAILS
TPS62913 uses 2.2 MHz, 470 nF NR/SS and 2.2 uH. Audio/headset variants add the optional ferrite second stage on their owning sheet.
6. AUX_12V BRANCHES AND AUDIO HANDOFF
DISPLAY_12V: 12 V / 2.5 A harness, simple fuse only. No dedicated monitor eFuse by requirement.
NIGHT_LIGHT_12V: independent 0.25 A fused panel branch; no CM5 or software dependency.
FAN_CPU_12V and FAN_AUX_12V are separate 3 A branches; every fan also has local resettable protection.
AUDIO-8X8 local: TRI 20-1223 creates +/-15 V. Separate TPS7A20 LDOs feed ADC 5 V, DAC 5 V and AKM 3.3 V; headset gets its own 1.8 V LDO.
7. SEQUENCING, POWER-GOOD, AND TEST ACCESS
RAW valid -> SYS_5V15, AUX_12V, LOGIC_3V3 and radio pre-regulators start from local UVLO.
SYS_5V15_PG enables LOGIC_1V8 and PCIE_1V0; hold PCIe reset until NET_3V3_PG and PCIE_1V0_PG are both valid.
WI-FI and modem final rails remain off until Thermal-IO asserts WIFI_POWER_EN / MODEM_POWER_EN after checks.
AUDIO_ENABLE controls the AUDIO-8X8 converters; DAC mute stays asserted through rail and clock qualification.
All PGOOD outputs are open drain; control-sheet 10k pull-ups to LOGIC_3V3 and adjacent rail test pads are required.
Component values are Rev A1 starting values; bench loop, thermal, and transfer validation remain ma
DISPLAY_12V is a fused 12 V / 2.5 A harness branch with no dedicated eFuse
22.4 mF / 50 V installed hold-up bank; optional stuffing raises total above 47 mF
RAW_OUT_LOAD accepts 10.5 V to 30 V after source selection and load telemetry
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